

Aaron Leevord Fernandes

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EDUCATION

Indiana University Bloomington

Aug 2023 – May 2025

M.S. in Intelligent Systems Engineering (Computer Engineering)

GPA: 3.9/4.0

- **Relevant Coursework:** Machine Learning, Deep Learning Systems, NLP, Reinforcement Learning, Image Processing, Cloud Computing

Goa University

Aug 2017 – Jul 2021

B.E. in Electronics and Telecommunication

GPA: 3.7/4.0

EXPERIENCE

Machine Learning Research Engineer (Computer Vision)— Leung Research Group, Indiana University Apr 2024 – Present

- Designed and evaluated an unpaired OCT-video → histology translation pipeline (CNN-GRU encoder-decoder + Transformer), reducing FID by **50%** vs a Fourier-domain baseline through controlled ablations.
- Implemented diffeomorphic registration (VoxelMorph-style), reducing frame-to-frame alignment error by **10%** (reconstruction MSE).
- Built scalable HPC synthetic data generation workflows (**Slurm + MPI**) for generating simulation data for metasurface design, producing **5K+** samples with **90%** wall-clock reduction.

Machine Learning Engineer (RL) — Indiana University (Concurrent Research Appointment)

Jan 2025 – June 2025

- Led development of a multi-goal sparse-reward RL pipeline (MuJoCo Robotics Environments), building evaluation tooling and reproducible experiment sweeps using a custom modular python codebase with **Hydra + W&B**.
- Improved success rate from **15%** → **55%** using HER; validated improvements through controlled replay and relabeling ablations.
- Implemented SVGG curriculum learning and evaluated curriculum strategies using averaged success rates across 100+ tracked runs.

Software Engineer — Tata Consultancy Services (HDFC Bank)

Jan 2022 – Jan 2023

- Contributed to the development of a microservices-based transaction processing system using Java, Spring Boot, Hibernate, and RESTful APIs, supporting 100K+ daily transactions in a production banking environment.
- Collaborated within a large cross-functional engineering team to develop and maintain production systems for a tier-1 banking client (HDFC Bank), following enterprise development and release processes.

SELECTED PROJECTS

Multimodal Text-Conditioned Image Generation (Stable Diffusion) — PyTorch, CLIP, DeepSpeed, DDP, ZeRO-2 Sept 2025 – Ongoing

- Implemented a **multimodal, promptable Stable Diffusion-style image generation pipeline**, conditioning a UNet denoiser on CLIP text embeddings via cross-attention for controllable text-to-image synthesis.
- Progressively developed the diffusion stack from baseline DDP and DDIM sampling to **latent diffusion** with **classifier-free guidance**, improving sampling efficiency and prompt adherence.
- Scaled training using **DeepSpeed** with distributed data parallelism and **ZeRO Stage-2** optimizer and gradient sharding, enabling efficient multi-GPU training of large diffusion models.

ComicPaliGemma – Visual Language Model — VLMs, LLMs, PyTorch

Dec 2025 – Ongoing

- Built a from-scratch implementation of Google's PaliGemma VLM (SigLIP ViT vision encoder + Gemma decoder), validating full pretrained checkpoint compatibility.
- Fine-tuned on a 5K+ comic-panel dataset using **QLoRA (4-bit)** to reduce peak GPU VRAM by **65%** while maintaining model quality.

Unsupervised Industrial Defect Detection — Anomaly Detection, ViT, PyTorch

Dec 2024 – Jan 2025

- Designed ViT-B/16 autoencoder with Gaussian Mixture Model (10 components) for anomaly scoring on BTAD dataset, achieving 20% AUROC improvement (0.94 vs 0.78) over ResNet-50 baseline

Grounded Theory Analysis Automation App — Electron.js, React.js, LangChain — NICC, Brussels

Oct 2024 - Dec 2024

- Developed a cross-platform desktop application integrating LLM-powered NLP pipelines (LangChain, LlamaIndex) with Knowledge Graphs (Neo4j) for scalable qualitative data analysis and relationship visualization.
- Targeted the 'coding' phase of GT framework, generating 90% of human codes and cutting analysis time from months to days.

Serverless Matrix Multiplication Using Map-Reduce — Google Cloud (GCP)

Aug 2024 - Sept 2024

- Formulated a serverless **Map-Reduce** matrix multiplication framework on **Google Cloud (Cloud Functions, Pub/Sub, Memorystore)** enabling scalable processing of matrices up to 10 GB, achieving a 25% efficiency gain by addressing key bottlenecks.

TECHNICAL SKILLS

ML Specializations: RL, Image Processing & Computer Vision, Diffusion Models, GenAI (LLMs, Finetuning, RLHF and RAG)

ML Frameworks & Libraries: PyTorch (incl. torch.profiler), TensorFlow, JAX, Triton, Ray, vLLM, Gymnasium (Mujoco), HuggingFace Transformers, Scikit-learn, NumPy, Pandas, Matplotlib, MLFlow, Weights & Biases

Cloud/HPC: AWS (SageMaker, EC2, S3, Lambda), GCP, Docker, Kubernetes, Slurm

Programming: Python (expert), C++ (CUDA/system programming), Java (Spring Boot), JavaScript/TypeScript(React.js/node.js), SQL, PostgreSQL, MongoDB, Neo4j, Redis, GraphQL, REST APIs

Certifications: [AWS Developer Associate](#), [AWS Cloud Practitioner](#)